

BAAVANSH REDDY GUNDLAPALLI

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SUMMARY

Computer Science graduate from Rutgers University (Magna Cum Laude, 3.76 GPA) targeting backend and AI/ML engineering roles. Builds REST APIs, authentication systems, and data pipelines, with applied AI work in retrieval and embeddings — including AgentMemry, an open-source memory library for AI agents. Backed by a strong systems foundation across CPU simulators, compilers, and operating systems, built in C and Python. Personal portfolio includes an AI assistant recruiters can chat with to ask about this work.

EDUCATION

Rutgers University — New Brunswick, NJ May 2026
B.S. in Computer Science, Magna Cum Laude | GPA: 3.76 / 4.0 | Dean's List (Fall '24, Spring '25, Fall '25, Spring '26)
Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Operating Systems, Compilers, Software Engineering, Databases, Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL
Frameworks & Web: Node.js, Express, FastAPI, React, Next.js, REST API design
Databases & Storage: PostgreSQL, MySQL, SQLite, Supabase, AWS S3, MinIO (S3-compatible object storage)
AI & ML: RAG pipelines, vector search, embeddings, semantic search, PyTorch, NumPy, prompt engineering, LLM evaluation
Cloud & DevOps: AWS (S3), Vercel, Docker, CI/CD pipelines, Git, GitHub, Linux
Tools & Concepts: unit testing (pytest, JUnit), object-oriented design, system design, REST API design, microservices concepts, Agile collaboration

EXPERIENCE

- Backend Developer Intern** • Vellarro AI Research Center (Metasys Global) Jan 2026 – Feb 2026
- Built the document-ingestion layer for an AI semantic-retrieval system, integrating MinIO (S3-compatible) object storage to programmatically extract heterogeneous, mixed-format files.
 - Engineered the file-processing workflow that staged and prepared extracted documents for downstream chunking, embedding, and vector search.
- Full-Stack Engineering Extern** • Fidelis — LIVEY Event Platform Jan 2026 – May 2026
- Co-built LIVEY, a full-stack event-discovery platform, by developing React search and filtering components and engineering REST API routes across the front and back end.
 - Integrated Supabase (PostgreSQL) for real-time data storage, user authentication, and event management across the platform.
- Software Engineering Extern** • Little Red Riding Hood Inc. Sep 2025 – Dec 2025
- Engineered secure authentication for a safety-focused rideshare platform, implementing role-based login and credential management for separate rider and driver accounts.
 - Built driver session tracking and real-time geolocation handling on AWS-deployed services, supporting dispatch coordination and passenger-safety workflows.
- Python Programming Intern** • InternPe Sep 2024 – Oct 2024
- Developed Python automation scripts for data processing and file handling, applying object-oriented design principles across structured engineering assignments.

PROJECTS

AgentMemry — Open-Source Memory Library for AI Agents | *Python, SQLite, sentence-transformers, pytest*

github.com/BaavanshReddy/agentmemry

- Built and open-sourced AgentMemry, a ~500-line Python library that gives AI agents persistent memory using local SQLite storage and on-device embeddings, with no cloud database or API keys required.
- Implemented semantic retrieval with cosine similarity and a Maximal Marginal Relevance (MMR) diversity mode, plus per-agent namespacing, and verified the public API with 15 passing unit tests.

Chat-with-Baavansh — AI Portfolio with Embedded RAG Assistant | *Next.js, TypeScript, Claude API*

baavansh-portfolio.vercel.app

- Built a personal portfolio site with an embedded RAG assistant that answers recruiter questions from a structured knowledge base, pairing a Claude-powered engine with an automatic in-browser fallback.

RISC-V CPU Simulator | *C, Computer Architecture*

github.com/BaavanshReddy/risc-v-simulator

- Implemented a single-cycle RISC-V CPU simulator with full instruction decode and datapath execution, backed by a 1KiB direct-mapped cache with hit/miss tracking to model a realistic memory hierarchy.

TinyL Compiler | *Python, Compilers*

github.com/BaavanshReddy/tinyl-compiler

- Built a complete compiler — lexer, recursive-descent parser, AST, intermediate representation, and code generation — translating the TinyL language from a formal grammar to stack-based bytecode.

RuPizza — Object-Oriented Ordering Application | *Java, JavaFX, JUnit*

github.com/BaavanshReddy/RUPizza

- Developed a JavaFX desktop ordering application designed with object-oriented principles — inheritance, polymorphism, and the Factory pattern — featuring an order builder, store-order management, and JUnit test coverage.

Neural Network from Scratch | *Python, NumPy, PyTorch*

github.com/BaavanshReddy/neural-net-from-scratch

- Implemented a perceptron and a 3-layer neural network with hand-coded forward propagation and backpropagation, reaching 89.1% face-recognition and 86.7% digit-classification accuracy validated against a PyTorch baseline.

Linux Filesystem Simulator | *C, Operating Systems*

github.com/BaavanshReddy/linux-filesystem

- Engineered a Unix-style virtual filesystem in C11 — inode-based storage, rwx permission management, and binary serialization for persistence — exposing 15+ shell commands through an interactive REPL and verified leak-free under Valgrind and AddressSanitizer.

Campus Event Scraper | *Python, BeautifulSoup, SQLite*

github.com/BaavanshReddy/campus-event-scraper

- Built a Python web-scraping pipeline that collects and normalizes university event listings into an indexed SQLite database, staying reliable through exponential-backoff retries, URL-based deduplication, and graceful offline fallback when sources are unavailable.

RESEARCH

LLM FactCheck — Factual QA Evaluation Framework | *Python, RAG, BM25, TriviaQA*

- Co-authored a research study benchmarking LLM factual question-answering across direct prompting, BM25 retrieval, and RAG on a controlled 100-question TriviaQA evaluation set.
- Found that BM25 surfaced supporting evidence in 90% of cases while RAG improved exact-match accuracy by only 3 points (0.66 to 0.69), isolating generation quality — not retrieval — as the bottleneck.

LEADERSHIP & ACTIVITIES

- **Community Service Officer, Rutgers University Police Department** (Jan 2025 – May 2026) — Supported incident response and crowd management for large-scale university events.
- **Leadership Roles, Alpha Phi Delta Fraternity** (May 2024 – May 2026) — Held three elected positions (Risk Management Chair, Head of Judiciary Board, Co-Brotherhood Chair); led safety compliance, conduct governance, and member engagement.
- **Administration Secretary, ENIGMA Technical Society** (Aug 2023 – May 2024) — Organized technical workshops and streamlined communication across society events.

Languages: English, Hindi, Telugu **Activities:** Former NCAA Division I Crew Athlete, Rutgers University